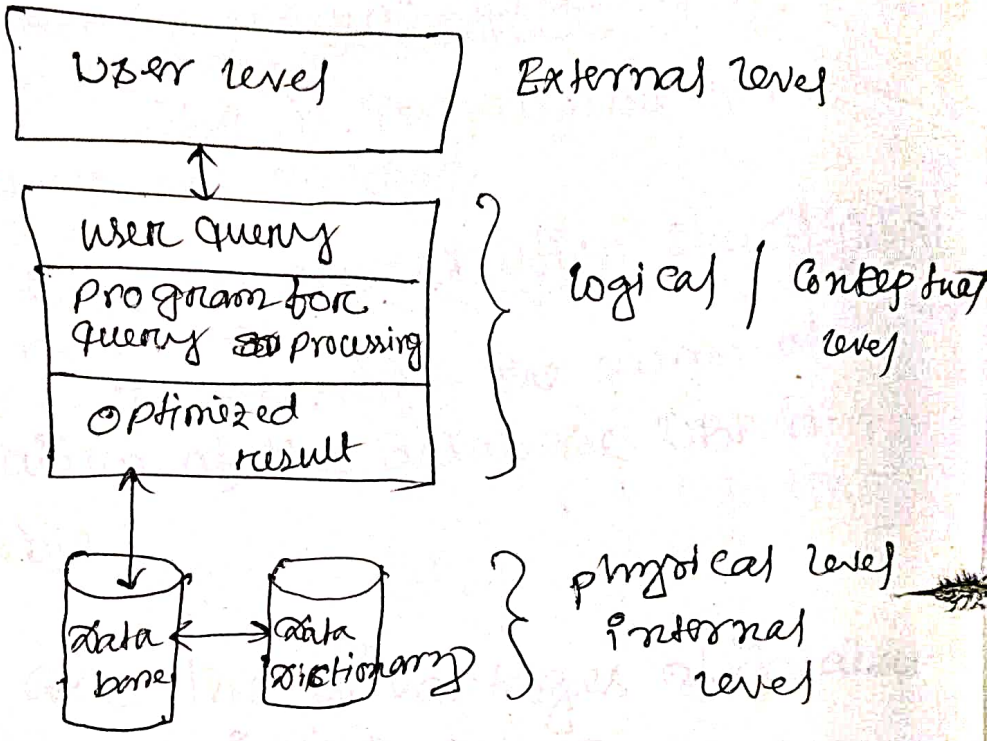
3 level architecture of DBMS / Data abstraction :

The entire DBMS can be divided into three layers. Hence three level architecture of DBMS is considered.



User level: This level is the outermost layer of DBMS. In this level the user write the required quary. It is also known as external level.

Logical level: In this level the quary obtained from external level are processed. This is also known as programing level.

Physical level: In this level there are two permanent storage - One is data base and another is data dictionary. The data dictionary contains the data about data, i.e, meta data.

When a data is stored within the database then related information about the stored data will be stored data dictionary and when the data will be retreived from the database then the corresponding definition of the data should be accessed from data dictionary.

Mapping: within the database two types of mapping are considered - One is external-logical mapping other is logical to physical mapping.

- External-logical mapping: It is the relation betn external and logical level of the DBMS.
- Logical-physical mapping: It is the relation betn logical and physical level of the DBMS.

Data independence:

The ability to modify the data in lower level without modifying the data in the next higer level is called data independance

Type: Two type of data independences are consider

(i) Physical data independence:

(ii) Logical data independence:

(i) Physical data independence:

The ability to modifying the data in the physical level without modifying the data in the logical level is called physical data independence.

(ii) Logical Data independence:

The ability to modify the data in the logical level without modifying the data at the user level is called logical data independence.

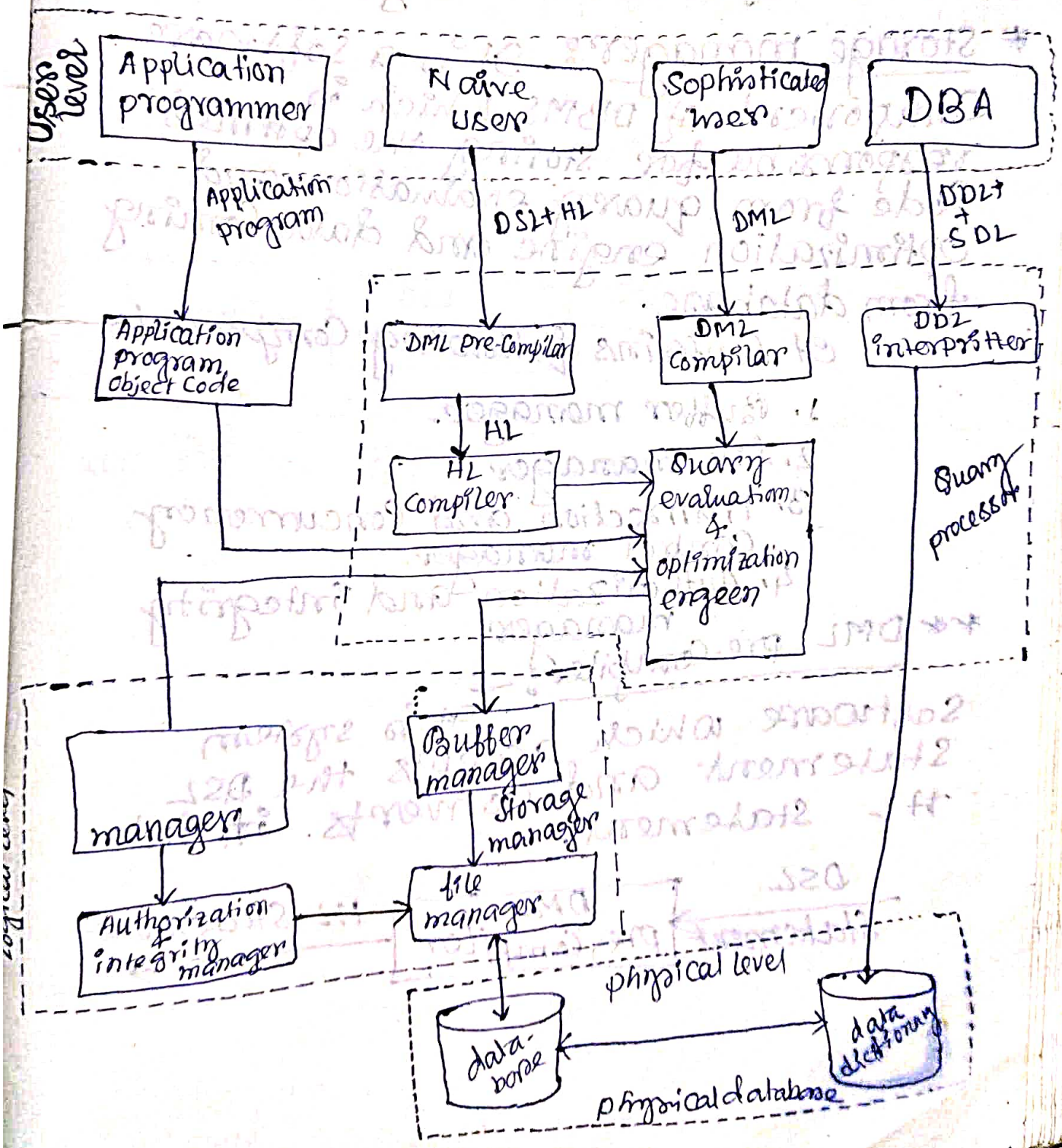
Disadvatage of DBMS:

(i) Centralized system: DBMS is the centralized system i.e, all the operation of an organization depend on the database when the database is crashed then the entire system will be postpond.

(ii) Expert user: To handle DBMS software all efficient user is needed.

(iii) Maintainence Cost: To implement and maintain database an efficient engeneer is needed. Hence the maintainence cost is higher as compare to file processing system.

Detailed architecture of DBMS:



Quary processor: It is a software i.e a component of DBMS which is used to processor contains following components

1. DML pre-compiler.
2. DML compiler.
3. DDL interpreter
4. HL compiler
5. Quary evaluation
6. Optimization engine

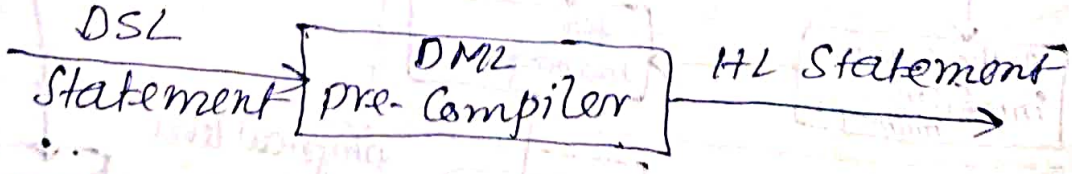
* **Storage manager:** It is a software component of DBMS which is responsible for storing the optimed code from quary evaluation and optimization engine and data coming from database.

It contains following components

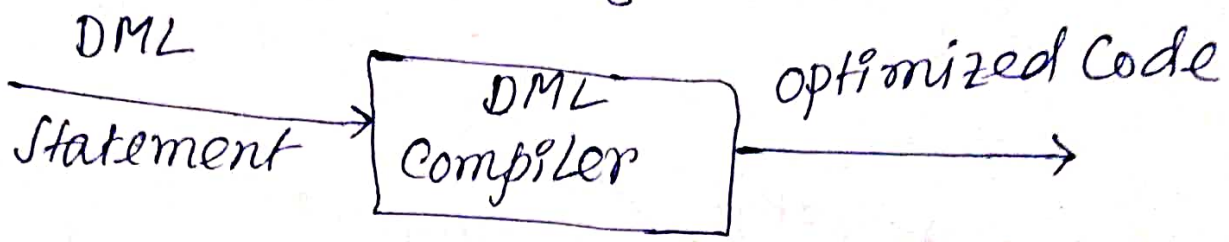
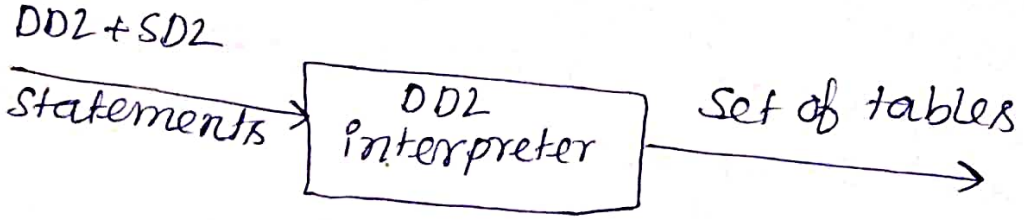
1. Buffer manager.
2. file manager
3. Transection and concurrency control manager.
4. Authorization and integrity manager.

** **DML pre-compiler:**

It is system software which accepts the DSL statement and converts it into HL statement.



** <u>DML Compilar</u> : It is a system Software which accepts the DML Statement and Converts it into optimized Code recognized by quarry evaluation and optimization ~~engine~~. engine



** <u>DDL interpreter</u>: It is a system S/w Which accepts the DDL & SDL Statement Converts it into Set of tables Stored Within the data dictionary.